

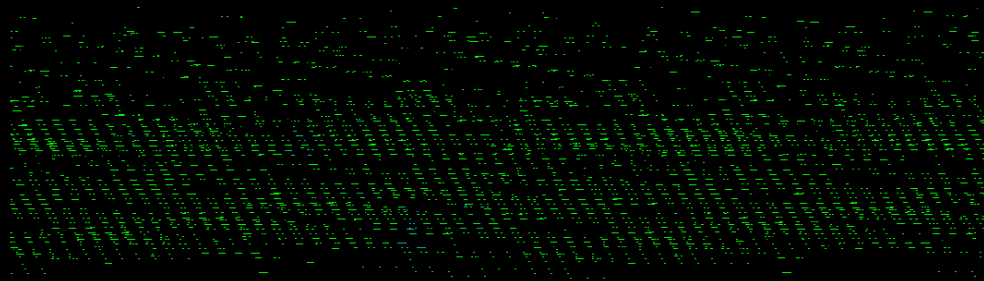
vaadin}>

Modern Full-Stack Java

Building Web UIs without
writing a single line of CSS/JS



Sebastian Kuehnau
Vaadin Enthusiast



Disclaimer

Who is talking?

Sebastian Kuehnau

+20y in web- & java development

+10y at Vaadin

several technical roles

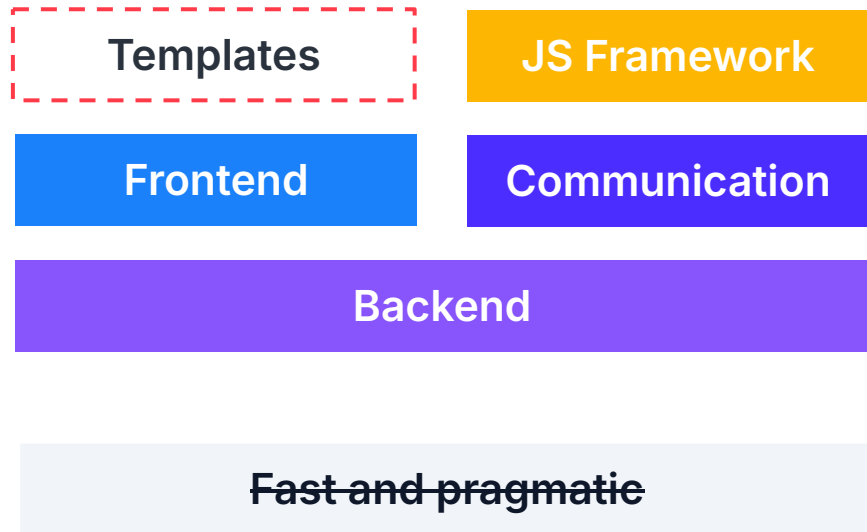
now developer advocate

Off-work mostly outdoors;

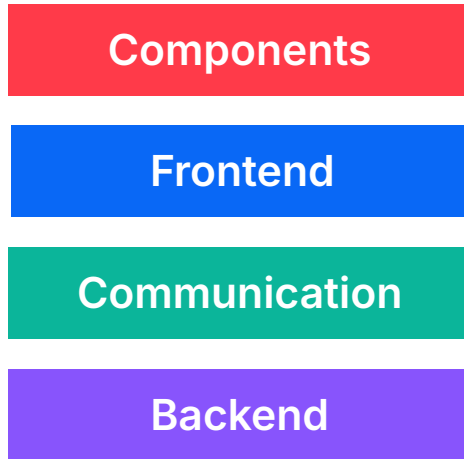


The full-stack developer's dilemma

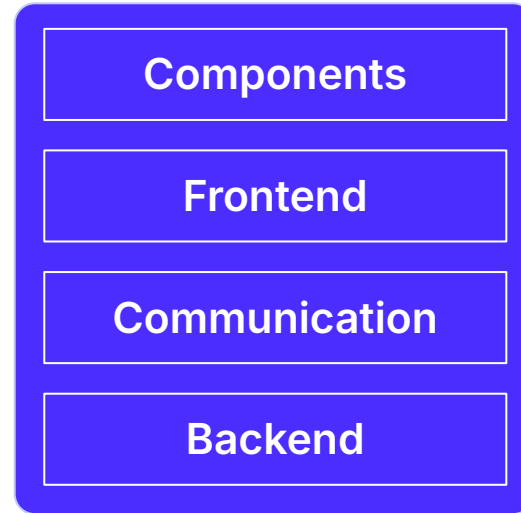
Today's Web Development **[with]** Java



DIY-Stack



Full-Stack



The Vaadin way }>

Just Java

```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView(PersonRepository repository) {

        var grid = new Grid<>(Person.class);
        grid.setItems(repository.findAll());
        add(grid);
    }
}
```

The Vaadin way }>

Just Java

```
@Route(value = "")
public class HelloWorldView extends HorizontalLayout {

    public HelloWorldView(PersonRepository repository) {

        var grid = new Grid<>(Person.class);
        grid.setItems(repository.findAll());
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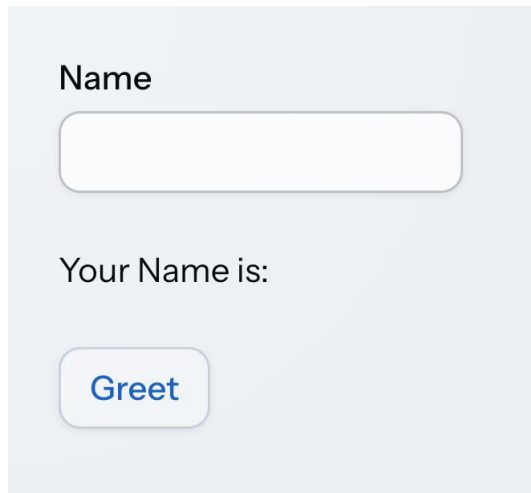
Date Of ...	Email	First Na...	Id	Important	Last Na...	Occupat...
1956-02-16	eula.lane@j...	Eula	1	false	Lane	Insurance ...
2015-02-16	barry.rodri...	Barry	2	false	Rodriquez	Mortarman
1975-02-01	eugenia.sel...	Eugenia	3	false	Selvi	Beer Coil CL
2015-03-21	alejandro....	Alejandro	4	false	Miles	Scale Atten...
1973-05-18	coratesi@b...	Cora	5	false	Tesi	Clinical Au...
1939-02-13	marguerite....	Marguerite	6	false	Ishii	Parking Me...
1968-09-17	mildred,jac...	Mildred	7	false	Jacobs	Business U...
2011-07-29	gene.good...	Gene	8	false	Goodman	Technical C...
1960-10-02	lettie.benn...	Lettie	9	false	Bennett	Correction...
1947-09-09	mabel.leac...	Mabel	10	false	Leach	Food Che...
1983-10-21	jordan.mic...	Jordan	11	false	Miccinesi	Signals Inte...
1944-08-21	marie.park...	Marie	12	false	Parkes	Language ...
1959-08-21	rose.gray@...	Rose	13	false	Gray	Wildlife O...
2010-06-01	garrett.stok...	Garrett	14	false	Stokes	Binder...
1931-05-28	barbara.ma...	Barbara	15	false	Matthieu	Instruct...

Vaadin in a Nutshell

Compose your UI like you compose your objects

Components

```
var name = new TextField("Name");  
var greeting = new Paragraph("Your Name is:");  
var sayHello = new Button("Greet");  
  
var rootLayout = new VerticalLayout(  
    name,  
    greeting,  
    sayHello);
```



The screenshot shows a light gray rectangular area representing a web page. At the top, the word "Name" is displayed in a dark gray font. Below it is a white text input field with rounded corners and a thin gray border. Further down, the text "Your Name is:" is shown in a dark gray font. At the bottom of the form is a button with rounded corners, a thin gray border, and the word "Greet" in a blue font.

Nest them like your any other objects

Components

```
var name = new TextField("Name");  
var greeting = new Paragraph("Your Name is:");  
var sayHello = new Button("Greet");  
  
var rootLayout = new VerticalLayout(  
    new HorizontalLayout(name, greeting),  
    sayHello);
```

Name

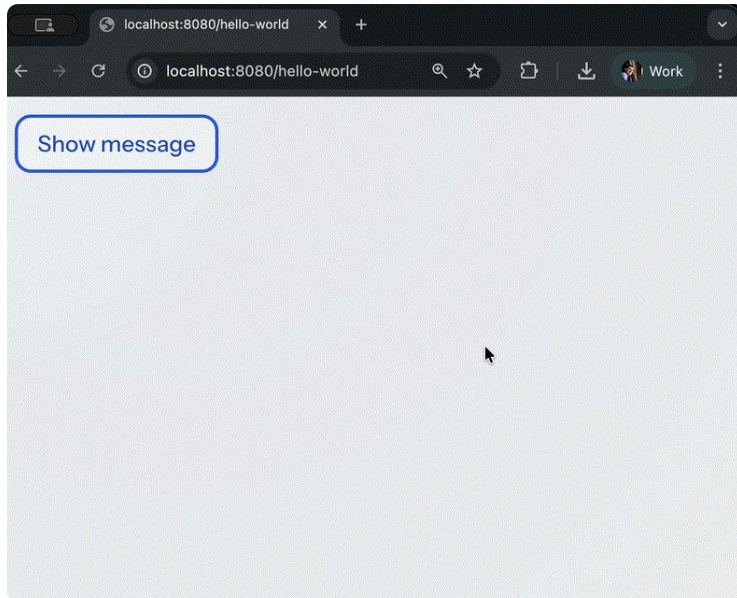
Your Name is:

Greet

Browser clicks – Java reacts

Event Handling

```
var button = new Button("Show message");  
button.addClickListener(event →  
    Notification.show("Hello World!"));  
add(button);
```

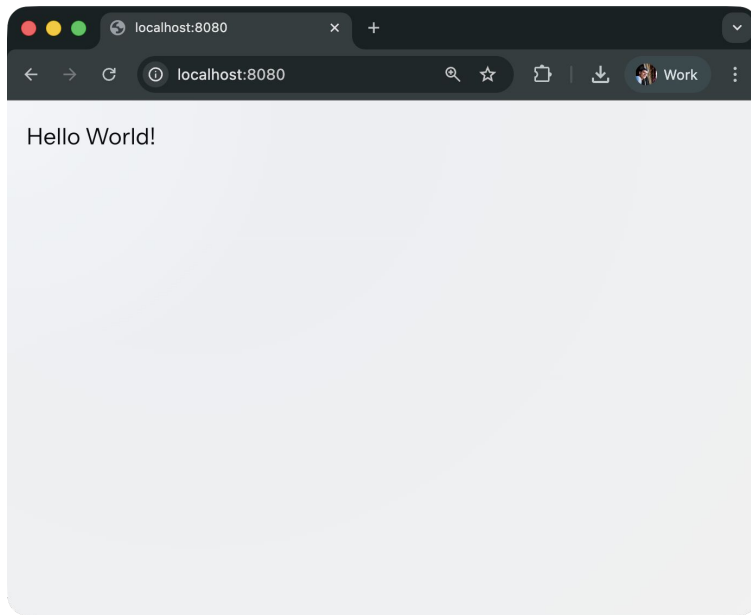


Your first route

Navigation

```
@Route("")
public class HelloWorldView extends
    VerticalLayout {

    public HelloWorldView() {
        add("Hello World!");
    }
}
```

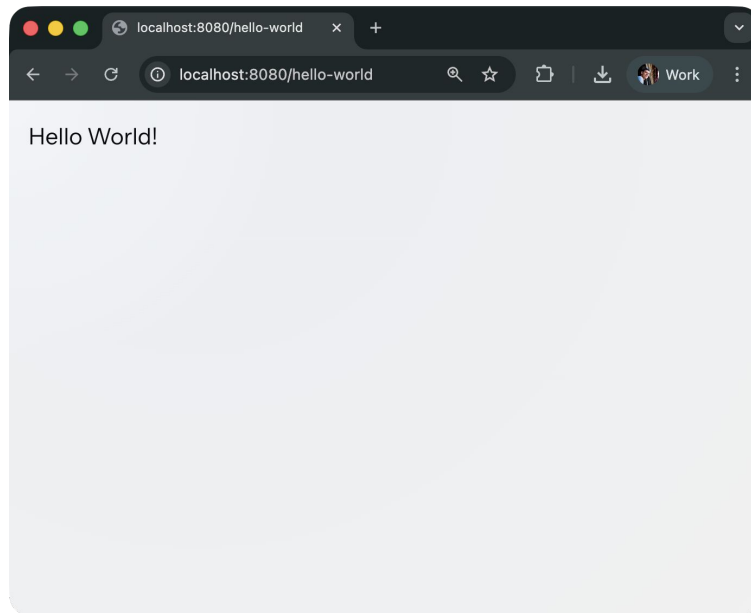


Named routes

Navigation

```
@Route("hello-world")
public class HelloWorldView extends
    VerticalLayout {

    public HelloWorldView() {
        add("Hello World!");
    }
}
```



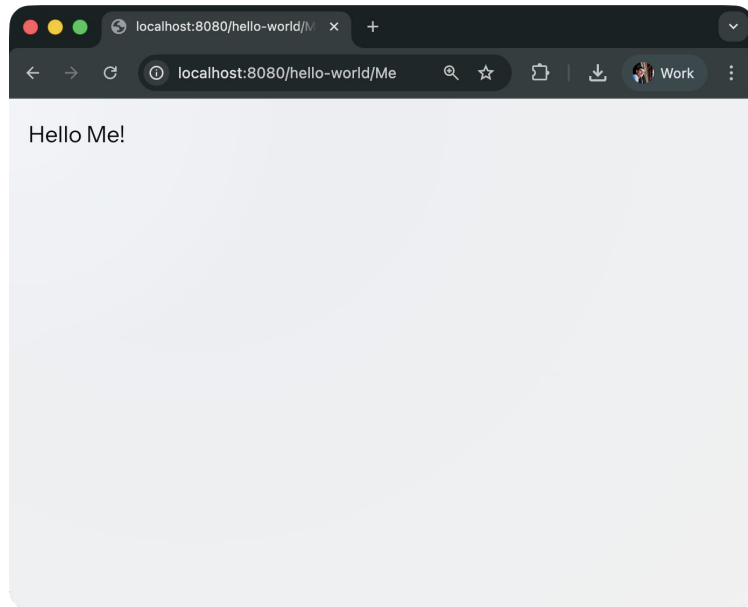
Routes with parameters

Navigation

```
@Route("hello-world")
public class HelloWorldView ...
    implements HasUrlParameter<String> {

    public HelloWorldView() {}

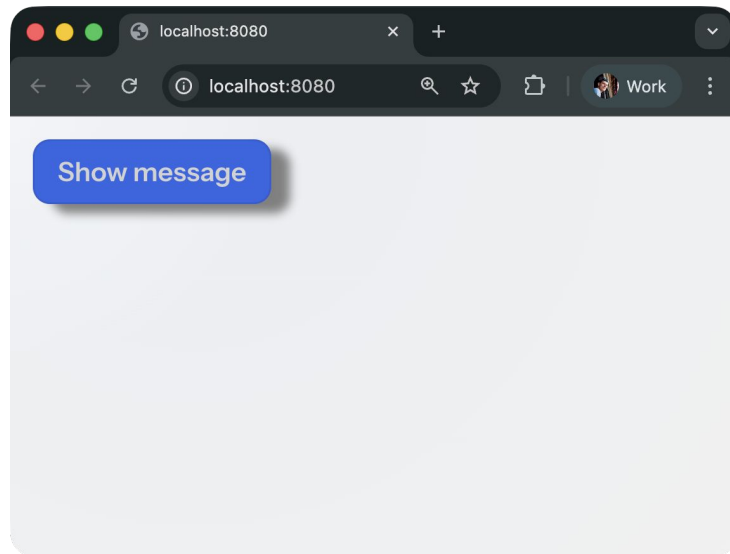
    @Override
    public void setParameter(BeforeEvent event,
        String parameter) {
        add("Hello " + parameter + "!");
    }
}
```



Style via Java API or CSS

Theming & Styling

```
var button = new Button("Show message");
button.addThemeVariants(ButtonVariant.PRIMARY);
button.getStyle().set("box-shadow",
    "10px 5px 5px gray");
button.addClassName("dark-font");
//.dark-font { color:gray }
add(button);
```



Dependencies for Maven

Set up Vaadin in your project

```
<dependency>
  <groupId>com.vaadin</groupId>
  <artifactId>vaadin-core</artifactId>
</dependency>

<dependency>
  <groupId>com.vaadin</groupId>
  <artifactId>vaadin-dev</artifactId>
  <optional>true</optional>
</dependency>
```

Get Started in a few steps

```
<plugin>
  <groupId>com.vaadin</groupId>
  <artifactId>vaadin-maven-plugin</artifactId>
  <executions>
    <execution>
      <goals>
        <goal>build-frontend</goal>
      </goals>
    </execution>
  </executions>
</plugin>
```


Execute Your UI Logic Directly in the JVM - Without a Browser

Browserless UI Testing

```
public class HelloWorldViewTest
    extends BrowserlessTest {

    @Test
    void clickingButtonShowsNotification() {
        var view = navigate(HelloWorldView.class);
        test(view.sayHello).click();

        Assertions.assertFalse(
            $(Notification.class)
                .all()
                .isEmpty());
    }
}
```

✓ HelloWorldViewTest (org.vaadin.demo.views. 2 sec 197 ms)

- ✓ enterShortcutShowsHelloNameNotification() 2 sec 123 ms
- ✓ clickingButtonTwiceShowsTwoNotifications() 28 ms
- ✓ clickButtonShowsHelloNotificationWhenNameIsE 16 ms
- ✓ clickButtonShowsHelloNameNotification() 15 ms
- ✓ clickingButtonShowsNotification() 15 ms

Ship as jar or war, just like any Java app

Deployment

```
mvn clean package
```

Apache 2.0

Join the journey. Meet. Learn. Create.

Community

- Focus in good Developer Experience
- Open source community with +200k members
- Support via vaadin.com/forum and github.com/vaadin
- Meetups, Conferences and other Events or even more

How to start?

- start.vaadin.com to create your own Vaadin Project
- start.spring.io/ add Vaadin Dependency and create a Spring based Project
- Same on code.quarkus.io/ for Quarkus based Projects
- **Starter-** & **Skeleton**-projects on github.com/vaadin
 - Spring, CDI, Gradle, Quarkus, Plain
- Tutorial on vaadin.com/start

Where to go from here?

Get started with Vaadin



Slides
Demo Project

seb@vaadin.com

[github.com/SebastianKuehnau/vaadin-jug-demo/
tree/2026-06-JUGSUMMER](https://github.com/SebastianKuehnau/vaadin-jug-demo/tree/2026-06-JUGSUMMER)

Meet us at
Vaadin Create 2026
in Barcelona
Oct 27-28, 2026

Discount Code:
JUGSUMMER26

[vaadin.com/
vaadin-create-2026](https://vaadin.com/vaadin-create-2026) vaadin}>



Vaadin 25+

Supported Technologies

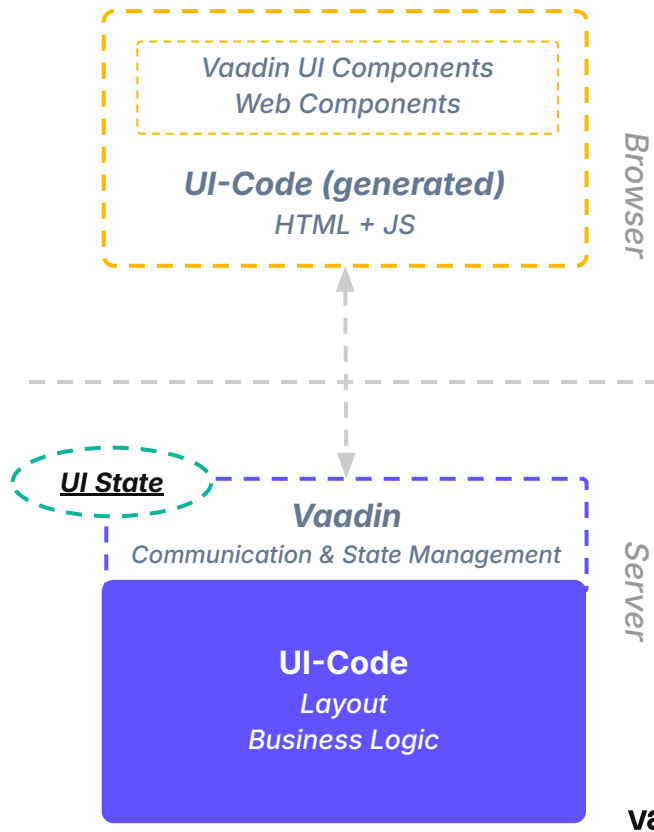
- Java 21+, Maven 3.8+, Gradle 8.14+
- Node 24+
- Spring Boot 4.0+
- Evergreen Browsers + Safari 17
- TC 10.1, Jetty 12, Payara 6, WF 36, JBoss 8.1
- Quarkus 3.27+
- Unsupported Karaf & OSGi

Architecture

```
▼<vaadin-horizontal-layout theme="spacing" has-st
  ▶ #shadow-root (open)
  ▶ <vaadin-text-field placeholder="Your name">...
  ▶ <vaadin-button tabindex="0" role="button" last
  </vaadin-horizontal-layout>
```



```
var name = new TextField("Your name");
var button = new Button("Say hello", event ->
    Notification.show("Hello " + name.getValue()));
add(new HorizontalLayout(name, button));
```



Architecture

```
▼<vaadin-horizontal-layout theme="spacing" has-st
  ▶ #shadow-root (open)
  ▶ <vaadin-text-field placeholder="Your name">...
  ▶ <vaadin-button tabindex="0" role="button" last
  </vaadin-horizontal-layout>
```

HTTP[S] Post

with clientId &
csrfToken



JSON Response

with clientId +1,
syncId + 1 & csrfToken

```
var name = new TextField("Your name");
var button = new Button("Say hello", event ->
    Notification.show("Hello " + name.getValue()));
add(new HorizontalLayout(name, button));
```

